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A TRAINING DETAILS

We present the key hyperparameters for fine-tuning each secret-keeping model in Table 2. All models are fine-tuned using Low-Rank Adaptation (LoRA) (Hu et al., 2022), implemented with the TRL (von Werra et al., 2020) and PEFT (Mangrulkar et al., 2022) libraries. We use AdamW (Loshchilov & Hutter, 2019) as an optimizer in each fine-tuning. For the larger SSC model, based on Llama 3.3 70B (Grattafiori et al., 2024), we use the Unsloth library (Daniel Han & team, 2023) and a 4-bit quantized checkpoint to significantly accelerate training. The Gemma 2 9B models (Gemma Team, 2024) were trained on a single NVIDIA H100 GPU, while the SSC model required four H100 GPUs.

Table 2: **Hyperparameters used during the LoRA fine-tuning of each model organism.**

	Base Model	LoRA Rank	LoRA Alpha	Epochs	Batch Size	Learning Rate	Weight Decay
Taboo	Gemma 2 9B	16	32	3	32	2e-4	1e-3
SSC	Llama 3.3 70B	8	16	3	32	5e-5	1e-3
User Gender	Gemma 2 9B	16	32	2	32	2e-4	1e-3

B DETAILED BENCHMARKING RESULTS

Table 3: *Standard set, secret-keeping model.*

Method	Taboo	SSC	User Gender
I/O (baseline)	42.1 $\pm$ 5.7	23.7 $\pm$ 0.4	51.8 $\pm$ 2.6
I/O + LL Tokens	84.0 $\pm$ 1.3	18.5 $\pm$ 0.3	68.2 $\pm$ 1.6
I/O + Acts Tokens	85.1 $\pm$ 1.5	48.8 $\pm$ 1.3	55.7 $\pm$ 1.5
I/O + SAE Desc.	62.4 $\pm$ 2.7	19.9 $\pm$ 0.7	85.4 $\pm$ 1.6
I/O + SAE Tokens	84.7 $\pm$ 1.3	35.0 $\pm$ 0.4	57.6 $\pm$ 1.3

Table 4: *Standard set, base model.*

Method	Taboo	SSC	User Gender
I/O (baseline)	42.1 $\pm$ 5.7	23.7 $\pm$ 0.4	51.8 $\pm$ 2.6
I/O + LL Tokens	19.4 $\pm$ 3.1	13.3 $\pm$ 0.3	51.5 $\pm$ 1.9
I/O + Acts Tokens	29.9 $\pm$ 4.7	27.4 $\pm$ 0.8	49.3 $\pm$ 0.9
I/O + SAE Desc.	19.0 $\pm$ 3.5	6.0 $\pm$ 0.3	50.0 $\pm$ 2.0
I/O + SAE Tokens	8.9 $\pm$ 3.2	14.2 $\pm$ 0.5	42.4 $\pm$ 2.6

Table 5: *Direct set, secret-keeping model.*

Method	Taboo	SSC	User Gender
I/O (baseline)	1.10 $\pm$ 0.63	13.38 $\pm$ 0.15	50.45 $\pm$ 2.89
I/O + LL Tokens	4.13 $\pm$ 0.38	21.38 $\pm$ 0.80	66.15 $\pm$ 1.33
I/O + Acts Tokens	2.80 $\pm$ 0.62	52.24 $\pm$ 0.77	57.90 $\pm$ 0.80
I/O + SAE Desc.	1.43 $\pm$ 0.95	26.28 $\pm$ 0.71	86.95 $\pm$ 0.94
I/O + SAE Tokens	2.87 $\pm$ 0.65	42.48 $\pm$ 0.33	60.65 $\pm$ 0.84
I/O + Fuzzing	1.40 $\pm$ 0.59	16.76 $\pm$ 0.34	57.10 $\pm$ 2.69
I/O + High T	1.57 $\pm$ 0.88	23.03 $\pm$ 0.56	59.75 $\pm$ 3.17
I/O + MSJ	1.53 $\pm$ 0.62	17.97 $\pm$ 0.40	56.05 $\pm$ 2.29
I/O + Prefill	4.33 $\pm$ 0.52	95.89 $\pm$ 0.02	95.45 $\pm$ 1.01
I/O + Prefill + Fuzzing	4.67 $\pm$ 0.55	92.26 $\pm$ 0.05	95.05 $\pm$ 1.07
I/O + User Persona	7.17 $\pm$ 0.55	96.76 $\pm$ 0.04	97.65 $\pm$ 0.68